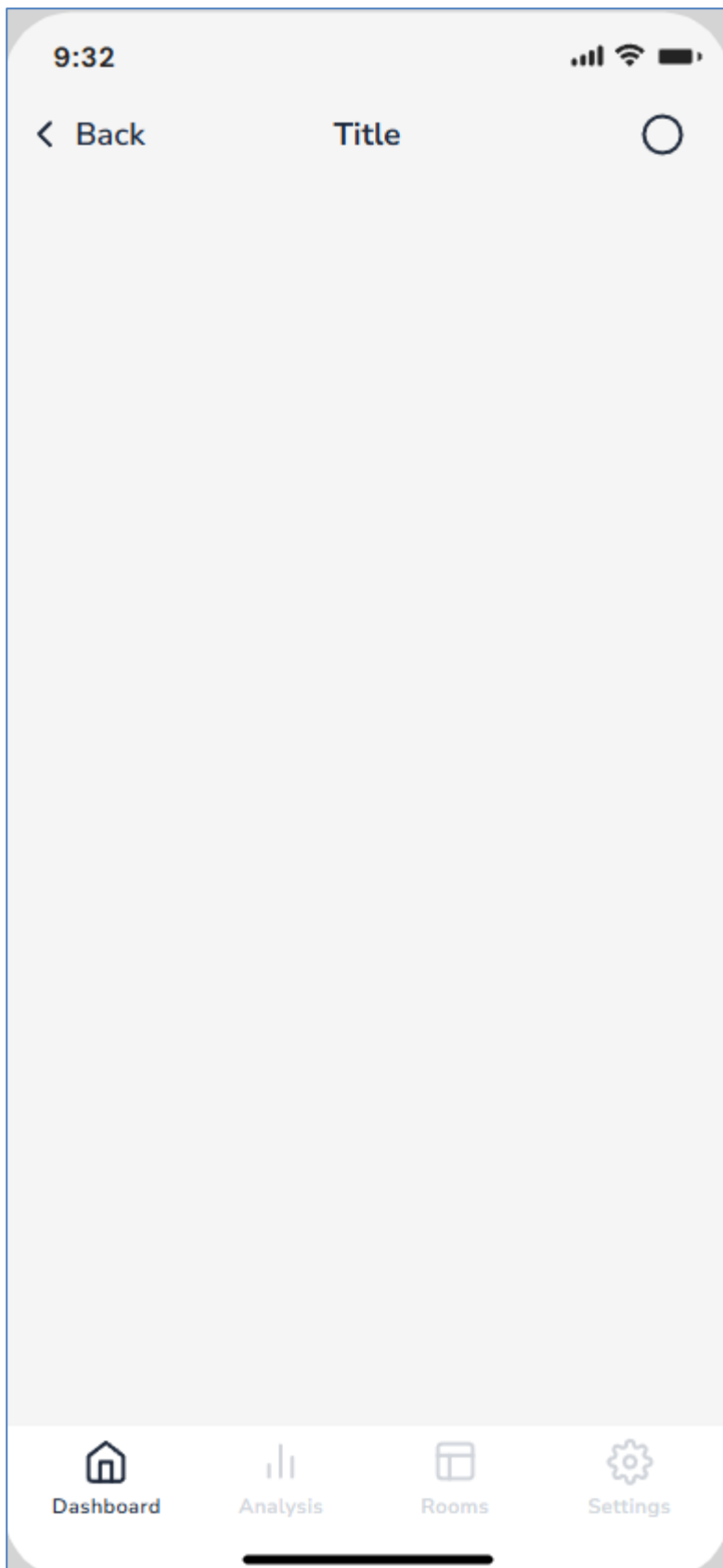


Exerciser 2.02

Mục tiêu: Tạo và quản App bật/tắt Led trên vi điều khiển **ESP32** qua giao thức **MQTT**

Yêu cầu: Ứng dụng thực hiện các chức năng sau:

- ✓ App quét QR Code để thêm thiết bị vào App
- ✓ Thực hiện bật/tắt LED trên board ESP32 từ Mobile app Android/iOS



Bước 0: Build app **ESP BLE Provisioning** từ source code:

✓ Android: <https://github.com/espressif/esp-idf-provisioning-android.git>

Bước 1: Thêm mới một **Empty View Activity**:

New Android Activity

Empty Views Activity
Creates a new empty activity

Activity Name
MainActivity

☒ Generate a Layout File

Layout Name
activity_main

☐ Launcher Activity

Package name
com.nguyenanhtu.exercise02

Source Language
Java

Cancel Finish

Bước 2: Tạo file **botom_nav_menu.xml** như sau:

```
res\menu\botom_nav_menu.xml
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/homeMenu"
        android:title="Home"
        android:icon="@drawable/ic_home"/>
    <item
        android:id="@+id/shopMenu"
        android:title="Shop"
        android:icon="@drawable/ic_shopping_cart"/>
    <item
        android:id="@+id/bagMenu"
        android:title="Bag"
        android:icon="@drawable/ic_bag"/>
    <item
        android:id="@+id/favMenu"
        android:title="Favorites"
        android:icon="@drawable/ic_fav"/>
    <item
        android:id="@+id/profileMenu"
        android:title="Profile"
        android:icon="@drawable/ic_profile"/>
</menu>
```

Bước 2: Sửa file **colors.xml** như sau:

```
res\values\colors\colors.xml
<?xml version="1.0" encoding="utf-8"?>
<resources>
    .
    .
    .
    <color name="primary">#DB3022</color>
```

```

<color name="background">#F9F9F9</color>
<color name="backgroundAd">#FFFFFF</color>
<color name="error">#F01F0E</color>
<color name="success">#2AA952</color>
<color name="sale">#DB3022</color>
<color name="black">#222222</color>
<color name="mainText">#222222</color>
<color name="textAd">#9B9B9B</color>
<color name="white">#FFFFFF</color>
<color name="purple_200">#FFBB86FC</color>
<color name="purple_500">#FF6200EE</color>
<color name="purple_700">#FF3700B3</color>
<color name="teal_200">#FF03DAC5</color>
<color name="teal_700">#FF018786</color>
</resources>

```

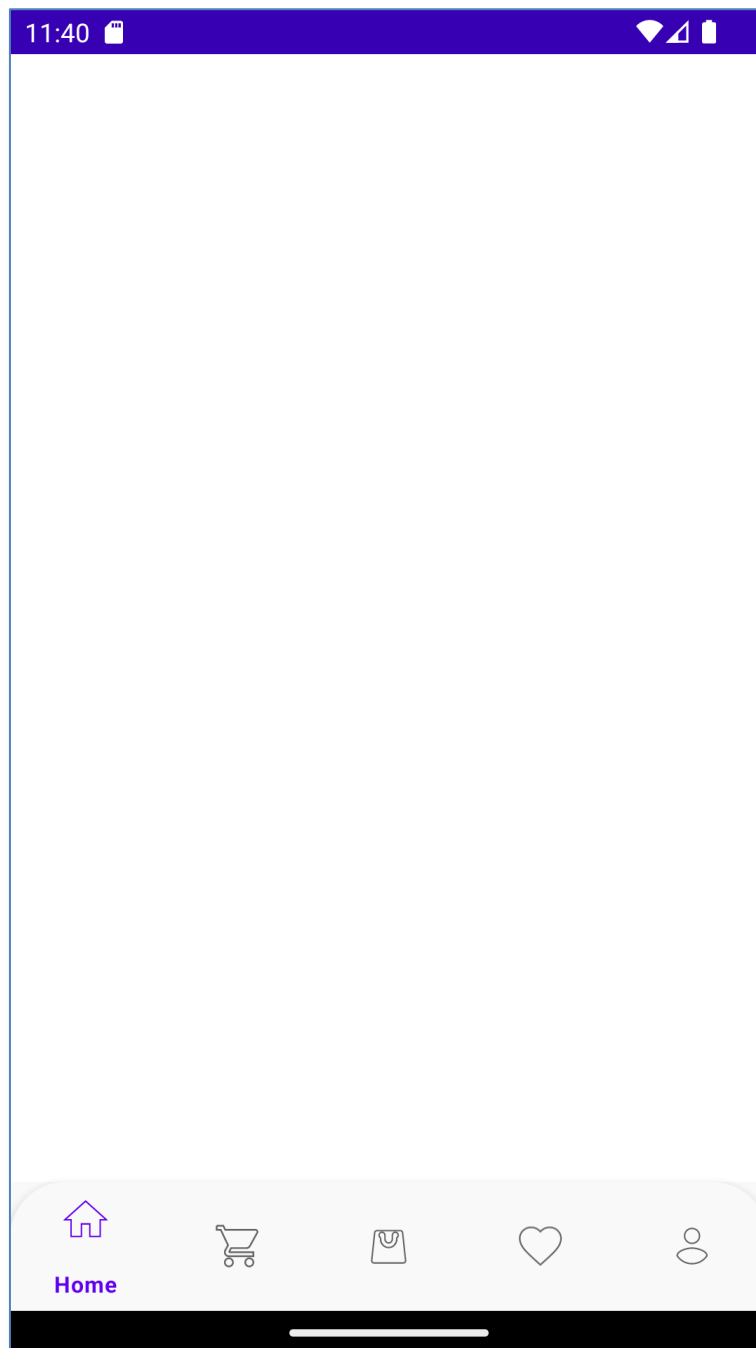
Bước 3: Sửa file **activity_main.xml** như sau:

```

res\layout\activity_main.xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:background="@color/background"
tools:context="com.nguyenanhtu.exercise02.MainActivity">
<FrameLayout
    android:background="@color/white"
    android:id="@+id/nav_fragment"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    app:layout_constraintBottom_toTopOf="@+id/bottomNavMenu"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent">
</FrameLayout>
<com.google.android.material.bottomnavigation.BottomNavigationView
    android:background="@drawable/menu_bg"
    android:id="@+id/bottomNavMenu"
    android:layout_width="match_parent"
    android:layout_height="70dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:menu="@menu/botom_nav_menu"
    />
</androidx.constraintlayout.widget.ConstraintLayout>

```

Bước 4: Run app (Shift + F10)



Bước 5: Tạo file **category_single.xml** như sau:

res\layout\category_single.xml
<pre><?xml version="1.0" encoding="utf-8"?> <androidx.cardview.widget.CardView xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" app:cardBackgroundColor="@color/backgroundAd" app:cardElevation="0dp" app:cardCornerRadius="10dp" android:layout_marginRight="5dp" android:layout_marginLeft="5dp" android:layout_marginBottom="5dp" android:layout_marginTop="5dp" android:layout_width="match_parent" android:layout_height="wrap_content"> <androidx.constraintlayout.widget.ConstraintLayout android:layout_marginBottom="10dp"></pre>

```

        android:layout_width="match_parent"
        android:layout_height="match_parent">
        <ImageView
            android:id="@+id/categoryImage_CateSingle"
            android:layout_width="match_parent"
            android:layout_height="70dp"
            android:scaleType="fitCenter"
            android:src="@drawable/iphone"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toTopOf="parent">
        </ImageView>

        <TextView
            android:id="@+id/categoryTitle_CateSingle"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginLeft="10dp"
            android:layout_marginRight="10dp"
            android:layout_marginTop="10dp"
            android:layout_marginBottom="10dp"
            android:fontFamily="@font/metropolis_regular"
            android:maxLines="2"
            android:text="Fashion Sale"
            android:textColor="@color/mainText"
            android:textSize="15sp"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toBottomOf="@+id/categoryImage_CateSingle"
        />
    </androidx.constraintlayout.widget.ConstraintLayout>
</androidx.cardview.widget.CardView>

```

Bước 6: Tạo file `fragment_home.xml` như sau:

res\layout\fragment_home.xml
<pre> <?xml version="1.0" encoding="utf-8" ?> <ScrollView xmlns:android="http://schemas.android.com/apk/res/android" xmlns:tools="http://schemas.android.com/tools" android:layout_width="match_parent" android:layout_height="match_parent" xmlns:app="http://schemas.android.com/apk/res-auto" android:background="@color/background" tools:context="com.nguyenanhtu.exercise202.MainActivity"> <androidx.constraintlayout.widget.ConstraintLayout android:id="@+id/MainLayout_homeFrag" android:layout_width="match_parent" android:layout_height="wrap_content"> <LinearLayout android:layout_marginTop="10dp" android:id="@+id/linearLayout12" android:layout_width="match_parent" android:layout_height="wrap_content" app:layout_constraintEnd_toEndOf="parent" app:layout_constraintStart_toStartOf="parent" app:layout_constraintTop_toTopOf="parent"> <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:layout_marginLeft="10dp" android:layout_marginRight="10dp"> <TextView android:layout_width="wrap_content" android:layout_height="match_parent" </pre>

```

        android:fontFamily="@font/metropolis_semibold"
        android:text="Phòng khách"
        android:textColor="@color/mainText"
        android:textSize="20sp" />
    <androidx.legacy.widget.Space
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1" />
    <TextView
        android:id="@+id/categories_GroupViewAll"
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:fontFamily="@font/metropolis_regular"
        android:gravity="center"
        android:text="Thêm thiết bị"
        android:textColor="@color/mainText"
        android:textSize="12sp" />
    </LinearLayout>
</LinearLayout>
<androidx.recyclerview.widget.RecyclerView
    android:layout_marginLeft="5dp"
    android:layout_marginRight="5dp"
    android:id="@+id/categoriesRecView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    app:layout_constraintEnd_toEndOf="parent"
    tools:listitem="@layout/category_single"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/linearLayout12" />
<LinearLayout
    android:layout_marginTop="10dp"
    android:id="@+id/linearLayoutHot"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/categoriesRecView">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginLeft="10dp"
        android:layout_marginRight="10dp">
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="match_parent"
            android:fontFamily="@font/metropolis_semibold"
            android:text="Phòng bếp"
            android:textColor="@color/mainText"
            android:textSize="20sp" />
        <androidx.legacy.widget.Space
            android:layout_width="0dp"
            android:layout_height="match_parent"
            android:layout_weight="1" />
        <TextView
            android:id="@+id/hot_GroupViewAll"
            android:layout_width="wrap_content"
            android:layout_height="match_parent"
            android:fontFamily="@font/metropolis_regular"
            android:gravity="center"
            android:text="Thêm thiết bị"
            android:textColor="@color/mainText"
            android:textSize="12sp" />
    </LinearLayout>

```

```
</LinearLayout>
<androidx.recyclerview.widget.RecyclerView
    android:layout_marginLeft="5dp"
    android:layout_marginRight="5dp"
    android:id="@+id/hotRecyclerView_shopFrag"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/linearLayoutHot" />
</androidx.constraintlayout.widget.ConstraintLayout>
</ScrollView>
```

Bước 7: Sửa file **build.gradle** như sau:

```
build.gradle

...
dependencies {
    implementation 'androidx.legacy:legacy-support-v4:1.0.0'
    ...
}
```

Bước 3: Tạo file HomeFragment.java như sau:**HomeFragment.java**

```
import android.app.Fragment;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

public class HomeFragment extends Fragment {
    @Override
    public void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
    }
    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
        return inflater.inflate(R.layout.fragment_home, container, false);
    }
}
```

Bước 4: Tạo file HomeActivity.java như sau:**HomeActivity.java**

```
package com.nguyenanhtu.exercise202;

import android.app.Fragment;
import android.app.FragmentManager;
import android.app.FragmentTransaction;
import android.os.Bundle;
import android.view.MenuItem;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

import com.google.android.material.bottomnavigation.BottomNavigationView;
import com.google.android.material.navigation.NavigationBarView;

public class MainActivity extends AppCompatActivity implements NavigationBarView.OnItemSelectedListener {
    BottomNavigationView bottomNavigationView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
    }
}
```

```
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);

bottomNavigationView = findViewById(R.id.bottomNavMenu);
bottomNavigationView.setOnItemClickListener(this);
loadFragment(new HomeFragment(), R.id.nav_fragment);
}

private void loadFragment(Fragment fragment, int layout) {
    FragmentManager fm = getFragmentManager();
    FragmentTransaction fragmentTransaction = fm.beginTransaction();
    fragmentTransaction.replace(layout, fragment);
    fragmentTransaction.commit();
}

@Override
public boolean onNavigationItemSelected(@NonNull MenuItem item) {
    if (item.getItemId() == R.id.homeMenu) {
        HomeFragment fragment = new HomeFragment();
        loadFragment(new HomeFragment(), R.id.nav_fragment);
    }
    return false;
}
}
```

Bước 5: Run app (Shift + F10)